The boundary diagonal is a sum of two complementary counts

$\mathrm{ord}_{S^\varepsilon_m}\,\widetilde{A}^X_{\mathbf{I},\mathbf{I}} = \#\{a \in M_m :\ q^{[m\notin\mathbf{I}]}\varepsilon^a = 1\}$,
and why a colength cannot tell (BD) from its complement

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Abstract

I answer Q544 and Q546. Both answers come from one theorem, which strictly contains the master formula of `2026-10-21-c2` (`thm:BD`) and has *no hypothesis on* q: for every specialised $q \in \mathbb{C}^\times$ and every $\varepsilon \in \mathbb{C}^\times$,

$$\mathrm{ord}_{S^\varepsilon_m}\,\widetilde{A}^X_{\mathbf{I},\mathbf{I}} \;=\; [\,m \in \mathbf{I}\,]\cdot f_M(\varepsilon) \;+\; [\,m \notin \mathbf{I}\,]\cdot g_M(q,\varepsilon), \qquad \begin{aligned} f_M(\varepsilon) &:= \#\{a \in M : \varepsilon^a = 1\},\\ g_M(q,\varepsilon) &:= \#\{a \in M : q\,\varepsilon^a = 1\},\end{aligned}$$

where $M = M_m(X) := \{a > 0 : a\,t_m \in \Delta^+(X_N)\}$ is the *multiplicity set* of the boundary direction. `lem:T`'s hypothesis $q \neq \varepsilon^{-h}$ *is* the statement that the second term vanishes; the composition point of Q544 is therefore not the formula's breakdown but the point where its suppressed term is maximal.

Q544: the two doublings do compose numerically, and the mechanism is a sum. At $(BC_N, \varepsilon = 1, q = 1)$ the order is 2 for *every* $\mathbf{I}$ and the colength is 2^{N+1}, so the pre-registered multiplicative guess is numerically correct. It is mechanistically wrong: the colength is $2^{N-1}(f_M + g_M)$, and the guess 2×2 agrees with the truth $2 + 2$ only by the arithmetic of 2. Neither "doubling" is a doubling. f is *additive* in M, so $f_{BC} = f_B + f_C$ and not $2f_B$ — at $\varepsilon = -1$ that reads $1 = 0+1 \neq 2\cdot 0$; and the q-degeneration leaves the $m \in \mathbf{I}$ half inert while switching on a complementary term. The first outright failure of multiplicativity is $M = \{1, 2, 4\}$ (3, against a guess of 4), covered by the theorem and realised by no root system.

A sharper consequence, pre-registered as F1 and confirmed. At $(B_N, \varepsilon = -1, q = -1)$ the order is $[\,m \notin \mathbf{I}\,]$ — the exact *complement* of (BD) — with colength 2^{N-1}, the *same integer* as (BD) itself. So **the colength cannot distinguish (BD) from its complement**: a flattening and an inversion are the same number.

Q546: false as asked, and the corrected form is a Möbius inversion. The ε-profile f_M at a primitive d-th root of unity is $\#(M \cap d\mathbb{Z})$, a sum-over-multiples transform of $\mathbf{1}_M$, so $[\,n \in M\,] = \sum_{k\geq 1}\mu(k) f_M(kn)$: the profile determines M_m *exactly and nothing more*. M_m is rank-blind, so B_2 and B_3 have equal profiles and the literal falsifier is satisfied. The *two*-point profile ($\varepsilon = \pm 1$) suffices for root systems only because $M_m \subseteq \{1, 2\}$ for every crystallographic system.

1 What was open

`2026-10-21-c2-G3-boundary-diagonal-typeB.tex` proves (`thm:BD`) that for $X \in \{B, C\}$,

$$\operatorname{ord}_{S^\varepsilon_m} \widetilde{A}^X_{\mathbf{I},\mathbf{I}} \;=\; [\, m \in \mathbf{I}\,] \cdot \#\{\, a > 0 :\; a\, t_m \in \Delta^+(X_N),\; \varepsilon^a = 1 \,\}, \tag{1}$$

under the hypothesis $q \neq \varepsilon^{-h}$ of its `lem:T`, which is automatic because q is transcendental over $\mathbb{C}$ in $F = \mathbb{C}(q)(w_1, \dots, w_N)$. Two questions were left open by that paper and sharpened by DREAM 10-21 c2.

Q544. Two separate "doublings" of the boundary colength were recorded. A non-reduced ambient (BC_N, in which t_m *and* $2t_m$ are positive roots) sends the root count $1 \to 2$, measured 49/49 at $N \leq 4$. Independently, $q \to 1$ at $\varepsilon = 1$ destroys the $\mathbf{I}$-dependence of (1) — the $m \in \mathbf{I}$ half is inert ($1 \to 1$) and the $m \notin \mathbf{I}$ half breaks ($0 \to 1$), measured 129/129, both types, $N \leq 5$ — taking the colength $2^{N-1} \to 2^N$. Do they compose? The point $(BC_N, \varepsilon = 1, q = 1)$ is $q = 1 = \varepsilon^{-h}$, *exactly* where `lem:T` fails, so (1) cannot be evaluated there and the natural guess 2^{N+1} had no derivation behind it.

Q546. The measured pairs (ord at $\varepsilon = 1$, ord at $\varepsilon = -1$) are $(1, 0)$, $(1, 1)$, $(2, 1)$ for B, C, BC and separate all three, while no single ε does. Is the pair a complete invariant of the ambient root system?

Both are answered by removing the hypothesis instead of working around it.

2 Setup: q becomes a scalar

I keep every convention of the 10-21 paper except one: q is specialised.

Write $M_m(X) := \{a > 0 : a\, t_m \in \Delta^+(X_N)\}$, the *multiplicity set* of the boundary direction t_m. For the reduced systems $M_m(B) = \{1\}$ and $M_m(C) = \{2\}$; for the non-reduced BC_N, $M_m(BC) = \{1, 2\}$. M_m does not depend on m, and I drop the subscript when convenient.

Definition 1 (the specialised valuation)**.** Fix $q_0 \in \mathbb{C}^\times$ and set $F := \mathbb{C}(w_1, \dots, w_N)$, $R := \mathbb{C}[w^{\pm}]$. Fix $m \in [N]$ and $\varepsilon \in \mathbb{C}^\times$. Since $R/(w_m - \varepsilon) \cong \mathbb{C}[w_j^{\pm} : j \neq m]$ is a domain, $(w_m - \varepsilon)$ is a prime of R; let $\mathcal{O} \subset F$ be the local ring, $t := w_m - \varepsilon$ the uniformiser, $k = \mathcal{O}/(t) \cong \mathbb{C}(w_j : j \neq m)$ the residue field, and ord_S the associated divisorial valuation on $F^\times$.

The only change from `def:ord` of 10-21 is that the coefficient field is $\mathbb{C}$ rather than $\mathbb{C}(q)$. Nothing in the construction used the transcendence of q: it used only that the w_j with $j \neq m$ are algebraically independent over the coefficient field, which they are over $\mathbb{C}$.

Lemma 2 (unit criterion, over $\mathbb{C}$)**.** *Let $M = \prod_j w_j^{a_j}$ be a Laurent monomial and $c \in \mathbb{C}^\times$. Then $1 - cM \in \mathcal{O}$, and*

(i) $\operatorname{ord}_S(1 - cM) \geq 1 \iff a_j = 0$ *for all $j \neq m$, and $c\,\varepsilon^{a_m} = 1$;*

(ii) *if moreover $a := a_m \neq 0$ then $\operatorname{ord}_S(1 - cM) = 1$ exactly, with cofactor $a\,\varepsilon^{a-1}$ at the generic point of S.*

Proof. Verbatim the proof of `lem:unit` of 10-21 with $\mathbb{C}(q)$ replaced by $\mathbb{C}$. Each w_j is a unit of $\mathcal{O}$ ($\varepsilon \neq 0$ for $j = m$), so $1 - cM \in \mathcal{O}$, with residue $1 - c\,\varepsilon^{a_m} \prod_{j \neq m} w_j^{a_j}$ in k. The w_j, $j \neq m$, are algebraically independent over $\mathbb{C}$, so this vanishes iff the monomial is constant and the scalar is 1, which is (i). For (ii), (i) gives $1 - cM = 1 - \varepsilon^{-a} w_m^a$; for $a > 0$, $1 - \varepsilon^{-a} w_m^a = -\varepsilon^{-a}(w_m - \varepsilon) \sum_{i=0}^{a-1} w_m^i \varepsilon^{a-1-i}$, and the sum has residue $a\varepsilon^{a-1} \neq 0$ in k because $\operatorname{char} \mathbb{C} = 0$ and $\varepsilon \neq 0$; for $a < 0$ factor out the unit $-\varepsilon^{-a} w_m^a$ and apply the case $-a > 0$. □

The single substantive point is that c **now ranges over a field containing** q_0. In the 10-21 argument $c \in \{1\} \cup q\mathbb{C}^\times$ with q transcendental, so $c\varepsilon^a = 1$ was *unsatisfiable* whenever c carried a q; that unsatisfiability is the whole content of `lem:T`, and it is exactly what specialisation removes.

3 The theorem

Recall from `eq:inv`, `eq:T`, `eq:diag` of the 10-21 paper that for $\mathbf{I} \subseteq [N]$ with strict partition $\lambda = \lambda(\mathbf{I})$,

$$\mathrm{Inv}(\lambda) = \{a\,t_k : k \in \mathbf{I},\ a \in M\} \cup \{t_j + t_k : j < k,\ j,k \in \mathbf{I}\} \cup \{t_k - t_j : j < k,\ j \notin \mathbf{I},\ k \in \mathbf{I}\}, \quad (2)$$

$$T(\lambda) = \{a\,t_k : k \notin \mathbf{I},\ a \in M\} \cup \{t_j + t_k : j < k,\ j,k \notin \mathbf{I}\} \cup \{t_k - t_j : j < k,\ j \in \mathbf{I},\ k \notin \mathbf{I}\}, \quad (3)$$

$$\widetilde{A}^X_{\mathbf{I},\mathbf{I}} = \prod_{\alpha \in \mathrm{Inv}(\lambda)} \left(1 - e^{-\alpha}\right) \prod_{\alpha \in T(\lambda)} \left(1 - q\,e^{+\alpha}\right). \quad (4)$$

For $X \in \{B, C\}$, M is a singleton and (2)–(4) are Motegi's theorem, read first-hand at L1305–1335 (type B) and L998–1046 (type C) of `arXiv:2610.15344`. For $X = BC$ there is no weight function in the literature: (4) with $M = \{1, 2\}$ is a *definition*, the non-reduced extension introduced as a lesion in `cor:red` of 10-21. Theorem 3 is therefore a theorem about Motegi's object when $X \in \{B, C\}$ and about my extension when $X = BC$; I flag this because the Q544 answer lives in the extension and only its mechanism is corroborated in the literature's types.

Theorem 3 (boundary diagonal at specialised q)**.** *Let $N \geq 1$, $\mathbf{I} \subseteq [N]$, $m \in [N]$, $M = M_m(X)$, and let $q_0, \varepsilon \in \mathbb{C}^\times$ be arbitrary. Then, with $q := q_0$ in (4),*

$$\mathrm{ord}_{S^\varepsilon_m} \widetilde{A}^X_{\mathbf{I},\mathbf{I}} \;=\; \#\big\{\, a \in M \;:\; q_0^{[\,m \notin \mathbf{I}\,]}\, \varepsilon^a = 1 \,\big\} \;=\; [\,m \in \mathbf{I}\,]\, f_M(\varepsilon) + [\,m \notin \mathbf{I}\,]\, g_M(q_0, \varepsilon), \quad (5)$$

where $f_M(\varepsilon) = \#\{a \in M : \varepsilon^a = 1\}$ and $g_M(q_0, \varepsilon) = \#\{a \in M : q_0\varepsilon^a = 1\}$. No hypothesis on q_0 is used.

Proof. ord_S is a valuation, so by (4) it is the sum of the orders of the finitely many factors, each of which lies in $\mathcal{O}$ by Lemma 2. Sort the factors by the support of the exponent vector of the monomial they contain.

Pair and difference factors are units, whatever q_0 is. For $\alpha = t_j \pm t_k$ with $j \neq k$ the exponent vector of $e^{\pm\alpha}$ is supported on two *distinct* indices, so some index $\neq m$ carries a nonzero exponent and Lemma 2(i) fails. This argument never inspects c, hence is untouched by specialisation. The same applies to boundary factors $a\,t_k$ with $k \neq m$, whose support is $\{k\} \not\ni m$.

The boundary factors at index m. By (2)–(3), for each $a \in M$ the root $a\,t_m$ lies in $\mathrm{Inv}(\lambda)$ if $m \in \mathbf{I}$ and in $T(\lambda)$ if $m \notin \mathbf{I}$ — in exactly one of the two, and the choice is the same for every $a \in M$. So the factors at index m are

$$\{\,1 - w_m^{-a} \;:\; a \in M\,\} \quad (m \in \mathbf{I}), \qquad \{\,1 - q_0\, w_m^{a} \;:\; a \in M\,\} \quad (m \notin \mathbf{I}),$$

one factor per $a \in M$. Lemma 2(i) applies with support $\{m\}$: the first has $c = 1$, $a_m = -a$, so its order is ≥ 1 iff $\varepsilon^{-a} = 1$, i.e. iff $\varepsilon^a = 1$; the second has $c = q_0$, $a_m = a$, so its order is ≥ 1 iff $q_0\varepsilon^a = 1$. In either case $a_m = \pm a \neq 0$, so Lemma 2(ii) makes the order *exactly* 1 whenever it is positive. Summing over $a \in M$ gives $f_M(\varepsilon)$ in the first case and $g_M(q_0, \varepsilon)$ in the second, which is (5). □

Remark 4 (`thm:BD` is the locus where the second term dies). If q is transcendental then $g_M(q,\varepsilon) = 0$, because $q\varepsilon^a = 1$ would make q algebraic; and $f_M(\varepsilon) = \#\{a > 0 : a\,t_m \in \Delta^+, \varepsilon^a = 1\}$ by definition of M. So (5) specialises to (1). More precisely, `lem:T`'s hypothesis $q \neq \varepsilon^{-h}$ is, for singleton $M = \{h\}$, *precisely* the condition $g_M(q,\varepsilon) = 0$. The hypothesis was not protecting the theorem from a degeneration; it was naming the locus on which one of its two terms vanishes. The 10-21 lesion ("$q := 1$ leaves the $m \in \mathbf{I}$ half inert and breaks the $m \notin \mathbf{I}$ half, 129/129") is (5) read at $q_0 = 1$: f does not mention q_0 at all, and $g_M(1,\varepsilon) = f_M(\varepsilon)$.

Remark 5 (the geometry: two families of hyperplanes, and the indicator chooses one). For each $a \in M$ the two boundary factors at index m have zero divisors

$$\{\, w_m^a = 1 \,\} = \textstyle\sum_{\zeta\in\mu_a}\{w_m = \zeta\}, \qquad \{\, q_0 w_m^a = 1 \,\} = \sum_{\eta^a = q_0^{-1}}\{w_m = \eta\},$$

the a-torsion of the w_m-line and a coset of it. All zeros are simple (Lemma 2(ii)). So there are $2\,|M|$ candidate loci through index m; the indicator $[\, m \in \mathbf{I}\,]$ selects a half of them, and the order counts how many of the selected ones pass through ε. The coset equals the torsion iff $q_0 = 1$, which is why $q_0 = 1$ makes the two halves agree for *every* ε and every X.

4 Q544: the composition point, and what "doubling" meant

Corollary 6 (the colength is a sum). *For every $q_0, \varepsilon \in \mathbb{C}^\times$,*

$$\operatorname{ord}_{S^\varepsilon_m} \det \Phi = \sum_{\mathbf{I}\subseteq[N]} \operatorname{ord}_S \widetilde{A}^X_{\mathbf{I},\mathbf{I}} \;=\; 2^{N-1}\big(f_M(\varepsilon) + g_M(q_0,\varepsilon)\big).$$

Proof. Exactly 2^{N-1} subsets contain m and 2^{N-1} do not; apply (5). □

Corollary 7 (Q544). *At $X = BC_N$, $\varepsilon = 1$, $q_0 = 1$: $M = \{1,2\}$ and $f_M(1) = g_M(1,1) = 2$, so* $\operatorname{ord}_S \widetilde{A}^{BC}_{\mathbf{I},\mathbf{I}} = 2$ *for* every $\mathbf{I}$ — *the order is constant in* $\mathbf{I}$ — *and* $\operatorname{ord}_S \det\Phi = 2^{N-1}\cdot 4 = 2^{N+1}$.

So the pre-registered multiplicative prediction is *numerically correct*, and the reason is worth stating precisely, because it is not multiplicativity.

Corollary 8 (neither "doubling" is a doubling). (a) *f_M is **additive in** M: $f_{M_1\sqcup M_2} = f_{M_1} + f_{M_2}$, immediately from the definition. Since $M(BC) = M(B)\sqcup M(C)$,*

$$f_{BC}(\varepsilon) = f_B(\varepsilon) + f_C(\varepsilon) \qquad \textit{and not} \qquad 2f_B(\varepsilon).$$

The two differ: at $\varepsilon = -1$, $f_B = 0$, $f_C = 1$, $f_{BC} = 1 = 0 + 1 \neq 2\cdot 0$. The whole measured $\{B, C, BC\} \times \{1, -1\}$ table is this one line — the BC row is the B row plus the C row — and "non-reducedness doubles the count" is true only at $\varepsilon = 1$, where $f_B = f_C$ by accident.

(b) *The q-degeneration does not multiply the order by anything. It leaves $f_M(\varepsilon)$, the $m \in \mathbf{I}$ half,* untouched *(f does not mention q_0), and switches on the complementary term g_M. At $q_0 = 1$ that term happens to equal f_M, which is why the* colength *doubles — $2^{N-1}(f+f) = 2\cdot 2^{N-1} f$ — and why it looked like a property of the order.*

(c) *Hence the composition law is $f + g$, and the agreement with the multiplicative guess at Corollary 7 is the coincidence $2\cdot 2 = 2 + 2$. The first outright failure is $M = \{1,2,4\}$ at $\varepsilon = -1$: the generic-q value is $f = 2$, the multiplicative guess for $q_0 = -1$ is 4, and the truth is $f + g = 2 + 1 = 3$. No root system realises $M = \{1,2,4\}$, so within the crystallographic family multiplicativity holds — for an arithmetic reason about $|M| \leq 2$, not a structural one.*

Corollary 9 (F1: the indicator inverts, and the colength cannot see it)**.** *At $X = B_N$, $\varepsilon = -1$, $q_0 = -1$: $M = \{1\}$, $f_M(-1) = 0$ and $g_M(-1,-1) = 1$, so*

$$\operatorname{ord}_{S_m^{-1}} \widetilde{A}^B_{\mathbf{I},\mathbf{I}} = [\, m \notin \mathbf{I}\,],$$

the exact complement *of* (BD)*, with colength 2^{N-1} — the* same integer *as (BD) itself. The same inversion occurs at $(C_N, \varepsilon = i, q_0 = -1)$ and at $(BC_N, \varepsilon = i, q_0 = -i)$.*

This is the sharp form of the Q544 finding, and it is the one I did not expect. The 10-21 paper's closing warning reads "a $q \to 1$ limit destroys (BD)'s dependence on $\mathbf{I}$ entirely". That is true at $q_0 = 1$ and is not the general phenomenon. The general phenomenon is that a second divisor family exists, is invisible while q is transcendental, and meets S^{ε}_m on a locus of its own; depending on where (q_0, ε) sits relative to the two families, the order is $[m \in \mathbf{I}]$, $[m \notin \mathbf{I}]$, constant, or zero. *Flattening is one case and inversion is another, and the colength 2^{N-1} is the same number for (BD) and for its complement.* Measured at $N = 3$, $m = 2$: both have colength 4, with supports $\{\mathbf{I} \ni 2\} = \{\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}$ and $\{\mathbf{I} \not\ni 2\} = \{\varnothing, \{1\}, \{3\}, \{1,3\}\}$ respectively.

Remark 10 (what this says about reporting)*.* Every statement of the form "the boundary colength of my lattice is 2^{N-1}" is a projection that forgets two things, not one: the divisor parameter ε (Q546's point) *and* the support of the order as a function of $\mathbf{I}$. The second loss is the worse one, because the support is what (BD) asserts. A report should carry the pair (f_M, g_M), from which the order, the support and the colength all follow.

5 Q546: the profile determines the multiplicity set, and nothing more

Theorem 11 (the ε-profile is a Möbius transform of $\mathbf{1}_M$)**.** *For $d \geq 1$ let ε_d be any primitive d-th root of unity and $f_M(d) := f_M(\varepsilon_d)$ — well defined, since $\varepsilon_d^a = 1 \iff d \mid a$ depends only on d. Then*

$$f_M(d) = \#\big(M \cap d\mathbb{Z}\big) = \sum_{n\,:\,d \mid n} \mathbf{1}_M(n), \qquad \textit{hence} \qquad \mathbf{1}_M(n) = \sum_{k \geq 1} \mu(k)\, f_M(kn) \quad (n \geq 1),$$

the sum being finite because M is finite. Consequently the profile $d \mapsto f_M(d)$ and the set M determine one another. For ε not a root of unity, $f_M(\varepsilon) = 0$, so the profile carries no further information.

Proof. The first display is the definition of f_M together with $\varepsilon_d^a = 1 \iff d \mid a$. The second is Möbius inversion for the sum-over-multiples transform: for $F(d) = \sum_{d \mid n} G(n)$ with G finitely supported, $\sum_{k \geq 1} \mu(k) F(kn) = \sum_{k \geq 1} \mu(k) \sum_{kn \mid n'} G(n') = \sum_{n \mid n'} G(n') \sum_{k \,\mid\, n'/n} \mu(k) = G(n)$, using $\sum_{k \mid j} \mu(k) = [\, j = 1\,]$. Finally if ε has infinite order then $\varepsilon^a \neq 1$ for all $a > 0$, so $f_M(\varepsilon) = 0$. □

Corollary 12 (Q546, as asked: **false**)**.** *The ε-profile — a fortiori the pair at $\varepsilon = \pm 1$ — is* not *a complete invariant of the ambient root system. By Theorem 11 it is a complete invariant of M_m, and M_m is rank-blind: $M_m(B_2) = M_m(B_3) = \{1\}$, so B_2 and B_3 have identical profiles. This is the pre-specified falsifier, and it is satisfied. The profile also cannot see the pair roots $t_k \pm t_j$ at all — by the proof of Theorem 3 they are units for every (q_0, ε) — so no amount of ε-data recovers them from this diagonal.*

Corollary 13 (Q546, corrected: **true**)**.** *The profile is a complete invariant of the multiplicity set* $M_m(X) = \{a > 0 : a\,t_m \in \Delta^+\}$*, i.e. of the set of positive roots proportional to* t_m*. Within* $\{B_N, C_N, BC_N\}$ *at fixed* N *the* two-*point profile* $(f_M(1), f_M(2)) = (1,0), (1,1), (2,1)$ *already separates the three, and this is not luck:* $f_M(1) = |M|$ *and* $f_M(2) = \#\{a \in M : 2 \mid a\}$ *determine* M *whenever* $M \subseteq \{1,2\}$*, and* ***every crystallographic root system has*** $M_m \subseteq \{1,2\}$ *— the only non-reduced ones are of type* BC*, where* α *and* 2α *are roots and* 3α *is not. So the sufficiency of two points is a fact about root systems, not about* $\widetilde{A}$*. For* $M \not\subseteq \{1,2\}$ *two points no longer suffice:* $M = \{1\}$ *and* $M = \{3\}$ *share* $(f(1), f(2)) = (1,0)$ *and are separated first at* $d = 3$*.*

Remark 14 (what the full profile costs). Reporting $(f_M(1), f_M(2))$ rather than one integer is the right repair *for this family*, but Corollary 9 shows it is still not enough on its own: the pair (f, g) — not the profile of f alone — is what determines the order and its support. At $(B_N, \varepsilon = -1, q_0 = -1)$ the f-profile is $(f(1), f(2)) = (1,0)$, identical to the generic-q type-B reading, while the order is the complement of (BD). The f-profile answers "which boundary roots resonate with ε"; it is silent about q_0.

6 Pre-registration: which arm fired

Written before computing, in `scratch/20261022-prove/PREREG.md`, and graded arm by arm. Seven falsifiers F1–F7 were pre-specified, *six of them away from* $q_0 = 1$, because the pre-registration already noted that $q_0 = 1$ cannot discriminate: there $g_M(1, \varepsilon) = f_M(\varepsilon)$ identically, so multiplicativity is vacuously true at the very point Q544 names.

Q544 numbers. Predicted 2 on $m \in \mathbf{I}$, 2 on $m \notin \mathbf{I}$, colength 2^{N+1}, order constant in $\mathbf{I}$: **all four TRUE** (Corollary 7).

The mechanism. Predicted the two-term sum (5) with $g = \#\{a \in M : q\varepsilon^a = 1\}$ and "$q \neq \varepsilon^{-h}$ is exactly $g = 0$": **TRUE** (Theorem 3, Remark 4). This is the arm that matters: the numerical prediction and the mechanism prediction were registered separately, and only the second distinguishes a sum from a product.

F1 (inversion in type B**), F2 (type** C**), F3 (**BC**).** Predicted ord $= [m \notin \mathbf{I}]$: **all three TRUE**, 34/34 each at $N \leq 3$ on the factor routes and, for F1 and F2, 34/34 on the independent symmetrisation route.

F4 (BC**,** $\varepsilon = 1$**,** $q_0 = -1$**):** $2[m \in \mathbf{I}\ .]$**TRUE**,34/34.**TRUE**,34/34.

F5 (BC**,** $\varepsilon = -1$**,** $q_0 = 1$**):** $\equiv$ **F6 (control,** B**,** $\varepsilon = i$**,** $q_0 = 2$**):** $\equiv 0$**. TRUE**, 34/34 — a genuine control, since here the prediction is 0 on both halves, so the formula is not fitted to the resonant points.

F7 (the Q544 point itself): $\equiv 2$**. TRUE**, 34/34.

Q546. Predicted "**NO** as stated, **YES** in the corrected form", with the literal falsifier satisfied by B_2 vs B_3 and the corrected statement a Möbius inversion over multiples: **TRUE on both halves** (Corollaries 12, 13).

Not pre-registered, and the session's actual find. That the colength is the *same integer* 2^{N-1} for (BD) and for its complement (Corollary 9). I registered the inversion; I did not register that it is invisible to the quantity this programme reports. The additivity $f_{BC} = f_B + f_C$ (Corollary 8(a))

was likewise not registered — I had written "non-reducedness doubles" in the pre-registration's own framing of the prior state, and the $\varepsilon = -1$ column of my own measured table had been contradicting it since 10-21.

7 Verification

Pure Python + SymPy (**SageMath is not installed**). q_0 and ε are exact SymPy numbers; $w_{j\neq m}$ stay *symbolic*, so every valuation is taken at the generic point of S and the codimension-2 hypersurfaces discussed in `rem:retract` of 10-21 cannot be hit. Scripts: `scratch/20261022-prove/{q544.py, routeC.py, q546.py, colength.py}`.

Three routes, of decreasing dependence on the objects under study.

Route A — factorwise. ord computed on each factor of (4) after $q := q_0$, and summed. Uses additivity of the valuation and the factor list.

Route B — whole product. The factors are multiplied first, $q := q_0$ substituted, and *one* valuation taken. Does not use additivity, so it is a check on the bookkeeping that Lemma 2 partitions.

Route C — symmetrisation. $\widetilde{A}^X_{\mathbf{I},\mathbf{I}}$ computed from Motegi's S_n-symmetrisation *definition* (`eq Motherfunction` L473–490, with $\mathcal{E}^B$ L1361–1367 / $\mathcal{E}^C$ L1057–1062), q kept symbolic, `cancel`led, and *then* specialised. Uses neither (4) nor the root system — the engine never mentions a positive root — so it is independent of everything Theorem 3 reasons about. Available for $X \in \{B, C\}$ only; BC has no symmetrisation, being defined by (4).

Domain check, pre-registered. The 10-19 lesson ("0/18 meant *undefined*, not false") was pre-registered here as an explicit check, since $q_0 = 1$ is a degeneration. Route C reduces $\widetilde{A}^X_{\mathbf{I},\mathbf{I}}$ to lowest terms as a rational function of q and then divides out every common factor of $(q - q_0)$ before substituting, reporting the count. **The count was** 0 **in every one of the** 7×34 **cases, and neither numerator nor denominator vanished at** q_0 **in any case.** So $q_0 = 1$ is in the domain: $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ is regular and nonzero there, and no 0/0 is being read as a number. This is as the theory predicts — (4) exhibits $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ as a *polynomial* in q, hence regular at every q_0 — but the prediction and the measurement are independent.

Measurements.

arm	(X, ε, q_0)	route A, $N \le 5$	route B, $N \le 3$	route C, $N \le 3$
F7 Q544 point	$(BC, 1, 1)$	258/258	34/34	— (BC)
F1 inversion, type B	$(B, -1, -1)$	258/258	34/34	34/34
F2 inversion, type C	$(C, i, -1)$	258/258	34/34	34/34
F3 inversion, BC	$(BC, i, -i)$	258/258	34/34	— (BC)
F4 asymmetric counts	$(BC, 1, -1)$	258/258	34/34	— (BC)
F5 $q_0 = 1$ off $\varepsilon = 1$	$(BC, -1, 1)$	258/258	34/34	— (BC)
F6 control, $\equiv 0$	$(B, i, 2)$	258/258	34/34	34/34
$q_0 = 1$, type B	$(B, 1, 1)$	258/258	34/34	34/34
$q_0 = 1$, type C	$(C, -1, 1)$	258/258	34/34	34/34
10-21 baseline B	$(B, 1, 3/7)$	258/258	34/34	34/34
10-21 baseline C	$(C, -1, 3/7)$	258/258	34/34	34/34
10-21 baseline BC	$(BC, 1, 3/7)$	258/258	34/34	— (BC)

The counts are all $(\mathbf{I}, m)$ pairs: $\sum_{N \le 3} 2^N N = 34$, and $34 + 64 + 160 = 258$ for $N \le 5$. Route A reaches $N = 5$ because each factor it inspects is a binomial; routes B and C cancel a product of $\Theta(N^2)$ factors and stop at $N = 3$. In every arm the measured order was *constant on each of the two halves* $m \in \mathbf{I}$, $m \notin \mathbf{I}$, as (5) requires; the two constants are (f_M, g_M) and are printed in the script output. The three baseline rows reproduce the 10-21 numbers on the same domain, so the specialised instrument agrees with the transcendental one where both apply.

Generalised multiplicity sets. (5) measured for M ranging over $\{1\}, \{2\}, \{1,2\}, \{3\}, \{1,3\}, \{2,3\}, \{1,2,3\}, \{2,$ — six of which no root system realises — against eight points (ε, q_0) including $(\zeta_3, 3/7)$, $(i, -1)$, $(i, -i)$, (ζ_3, ζ_3^{-1}): 2720/2720 at $N \le 3$. This is the measurement that bears on Theorem 11, whose content is the map $M \mapsto f_M$.

Colength. Corollary 6 measured by summing the order over all $\mathbf{I}$: 60/60 over ten (M, ε, q_0) points at $N \le 3$, including $M = \{1, 2, 4\}$ at $(-1, -1)$, where the measured colength is $3 \cdot 2^{N-1}$ against the multiplicative guess $4 \cdot 2^{N-1}$ — Corollary 8(c).

The support. At $N = 3$, $m = 2$: $(B, 1, 3/7)$ gives colength 4 with support $\{\mathbf{I} \ni 2\}$; $(B, -1, -1)$ gives colength 4 with support $\{\mathbf{I} \not\ni 2\}$. Equal integer, complementary support.

Möbius recovery. $\mathbf{1}_M(n) = \sum_k \mu(k) f_M(kn)$ recovered M exactly in 11/11 cases (the ten above plus $\{4, 6\}$), with the profile computed from *explicit* primitive roots of unity $e^{2\pi i/d}$, $d \le 12$, not from the divisibility rule — so the identification $\varepsilon_d^a = 1 \iff d \mid a$ is measured rather than assumed.

Remark 15 (instrument plumbing, recorded). The trajectory emitter was initialised with an *absolute* `log_dir` and a *stable* session tag `Q544-20261022`, which is the 10-18 repair: the default

timestamp session fragmented one session into four files while `report` reads one. A first call made before the repair wrote a bare-timestamp file; it was deleted and the event re-emitted into the stable sink rather than left as a second fragment. 18 events emitted, 18 present in the report — counted, not assumed, since a `verify` lacking `verification_target` is silently dropped.

I also corrected one of my own encodings, unprompted by the checker. The Q546 claim listed the generalised-M computation in its `inputs`, while the `verify` event for that claim offered the *same* computation as evidence — so the verification's evidence was an ancestor of its own target. This is the 2026-10-18 fault class (reading `inputs` as "what this is *about*" rather than "what this *depends on*"), and it is its fifth instance. Theorem 11 was in fact proved by hand from Theorem 3; the computation came afterwards and is evidence, not a dependency. Re-emitted with `inputs` corrected; the as-recorded events are retained in the same file rather than deleted.

8 What this closes, and what it does not

Q544 — CLOSED. Corollary 7: order $\equiv 2$, colength 2^{N+1}, constant in $\mathbf{I}$. The two doublings compose as a *sum* $f + g$, and the multiplicative guess is right at this point only because $2 \cdot 2 = 2 + 2$ (Corollary 8).

Q546 — CLOSED, both halves. False as asked (Corollary 12); true as the statement that the profile determines M_m (Corollaries 13, Theorem 11).

`thm:BD` **— strictly generalised, hypothesis removed.** Theorem 3 has no hypothesis on q and contains (1) as the locus $g_M = 0$. Nothing in the 10-21 paper is retracted; one of its closing warnings is *narrowed* — "a $q \to 1$ limit destroys (BD)'s dependence on $\mathbf{I}$" is the $q_0 = 1$ case of a family that also inverts the dependence (Corollary 9).

A caveat on scope. For $X = BC$ the object is defined by (4); route C does not apply and the BC arms rest on routes A and B. The *mechanism* — the two complementary blocks, and the inversion — is confirmed on the independent symmetrisation route in types B and C, which are Motegi's.

Unchanged. G2 (non-cancellation at fixed χ) is untouched; nothing above bears on it. The shared unverified RTV stable-envelope root beneath (4) is not reduced by this work — Theorem 3 takes (4) as its input exactly as 10-21 did.

Downstream obligation. Every colength in §§5–7 of `2026-10-19-c2-sa-divisors-full-q.tex` is stated at transcendental q and so sits at $g_M = 0$; those statements stand. What must not be inherited is the *form* 2^{N-1}: under any specialisation of q the correct report is the pair (f_M, g_M), which determines the order, its support, and the colength, and which no single integer determines.